HDOC Day-45

Menu-Driven C Program Using Separate Functions

Aim: To write a menu-driven C program that (1) converts a date from dd/mm/yyyy to dd-mmm-yy format and (2) prints all substrings of a given string, using separate user-defined functions.

Option	Task	Example
Operation 1	Date conversion	Input: 12/10/2026 -> Output: 12-Oct-26
Operation 2	Print all substrings	Input: abc -> Output: a, ab, abc, b, bc, c

Algorithm

1. Start the program and display a menu with three choices: date conversion, substring generation, and exit.

2. If the user selects date conversion, read day, month, and year in dd/mm/yyyy format.

3. Validate the month and convert its number to the corresponding three-letter month name.

4. Print the date in dd-mmm-yy format using the last two digits of the year.

5. If the user selects substring generation, read a string.

6. Use nested loops: choose each starting position, then print every possible ending position from it.

7. Repeat the menu until the user selects Exit.

C Program

```
#include <stdio.h>
#include <string.h>

void changeDateFormat(void);
void printSubstrings(void);

int main(void)
{
   int choice;

   do
   {
      printf("\n--- MENU ---\n");
      printf("1. Change date format\n");
      printf("2. Print all substrings of a string\n");
      printf("3. Exit\n");
      printf("Enter your choice: ");
      scanf("%d", &choice);
```

```
        switch (choice)
        {
            case 1:
                changeDateFormat();
                break;
            case 2:
                printSubstrings();
                break;
            case 3:
                printf("Program terminated.\n");
                break;
            default:
                printf("Invalid choice. Please try again.\n");
        }
    } while (choice != 3);

    return 0;
}

void changeDateFormat(void)
{
    int day, month, year;
    const char *months[] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};

    printf("Enter date in dd/mm/yyyy format: ");
    if (scanf("%d/%d/%d", &day, &month, &year) != 3 || month < 1 || month > 12)
    {
        printf("Invalid date format.\n");
        return;
    }

    printf("Converted date: %02d-%s-%02d\n",
           day, months[month - 1], year % 100);
}

void printSubstrings(void)
{
    char str[100];
    int i, j, k, n;

    printf("Enter a string: ");
    scanf("%99s", str);
    n = strlen(str);

    printf("Substrings: ");
    for (i = 0; i < n; i++)
    {
        for (j = i; j < n; j++)
        {
            for (k = i; k <= j; k++)
                printf("%c", str[k]);

            if (!(i == n - 1 && j == n - 1))
                printf(", ");
        }
    }
```

```
    printf("\n");
}
```

Sample Test Cases

Test Case	Input	Expected Output
1	Choice: 1 12/10/2026	12-Oct-26
2	Choice: 2 abc	a, ab, abc, b, bc, c
3	Choice: 3	Program terminated.

Functions Used

Function	Purpose
main()	Displays the menu and controls program execution.
changeDateFormat()	Reads a date and converts the numeric month to a three-letter month name.
printSubstrings()	Generates and prints every substring of the entered string.
strlen()	Finds the length of the input string.

Header Files Used

#include <stdio.h> - used for printf() and scanf().

#include <string.h> - used for strlen().

Result

The menu-driven C program was written using separate functions to convert the date format and to print all substrings of a string.